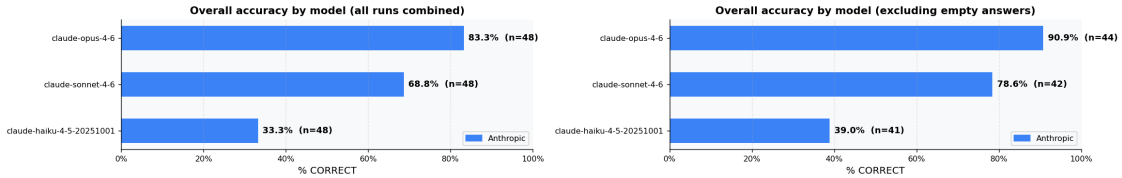
results_analysis_v2

March 13, 2026

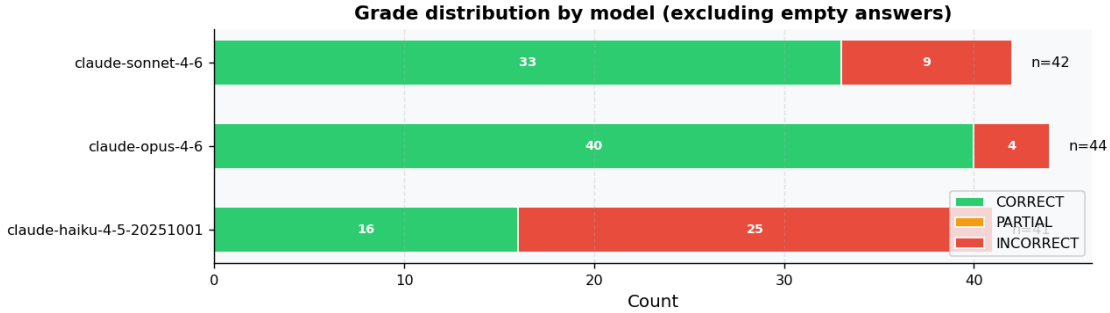
1 TinyBench — Evaluation Results Analysis

```
Loaded 2 result file(s):
  ../data/results/graded/eval_python_only_graded.json  →  run='python_only'
  ../data/results/graded/eval_python_semantic_graded.json  →  run='semantic'

144 total records  |  3 models  |  2 runs  |  2 templates
Empty answers: 17
Records after excluding empty answers: 127 / 144
Runs present: ['python_only', 'semantic']
```

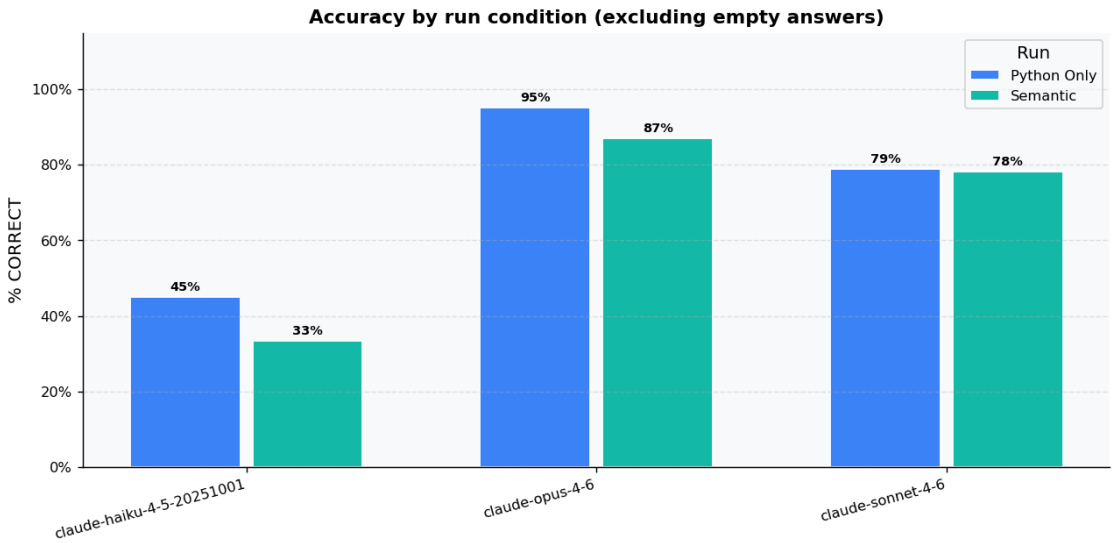
1.1 1 — Overall Accuracy by Model



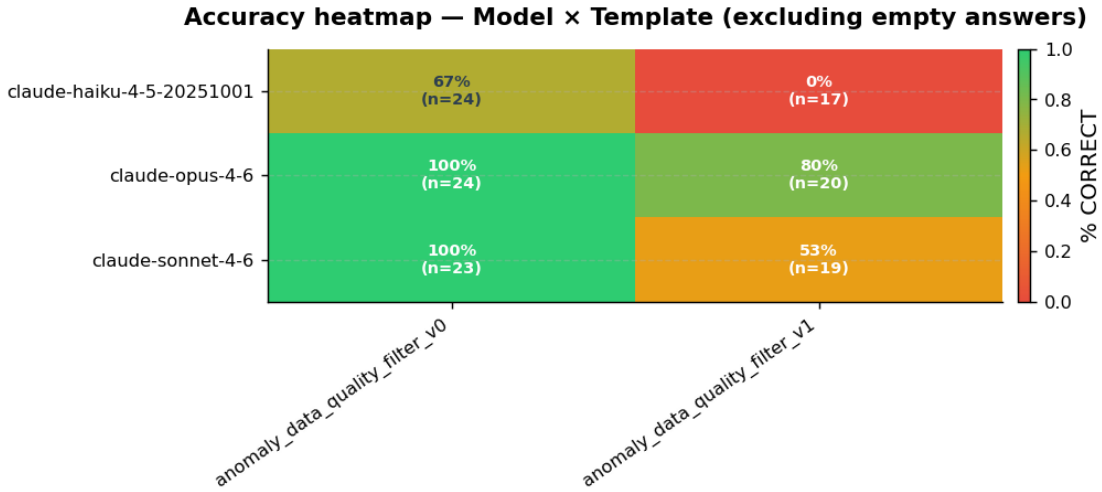
1.2 2 — Grade Distribution



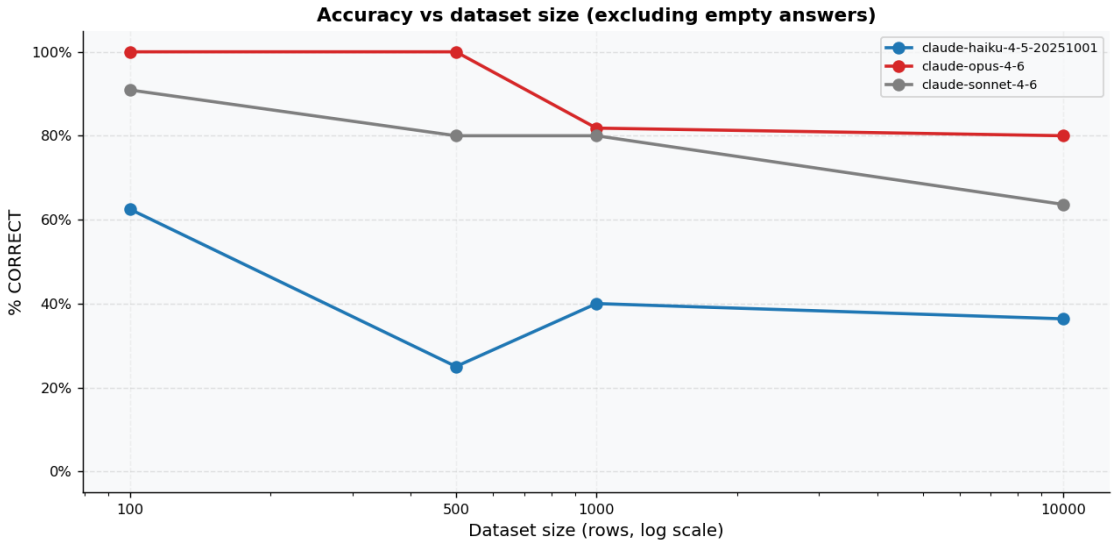
1.3 3 — Accuracy by Run Condition



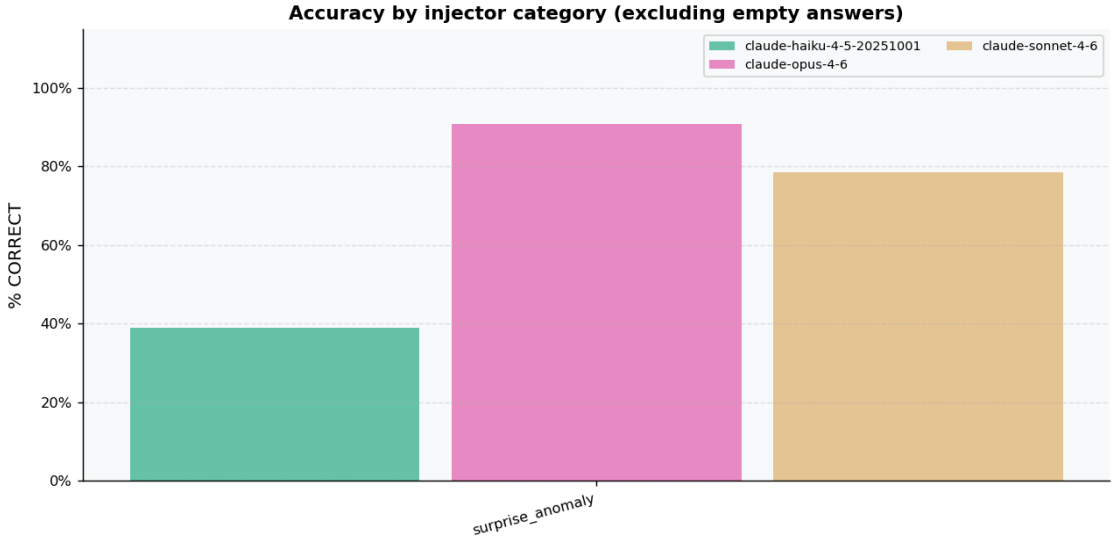
1.4 4 — Accuracy Heatmap: Model × Template



1.5 5 — Accuracy by Dataset Size



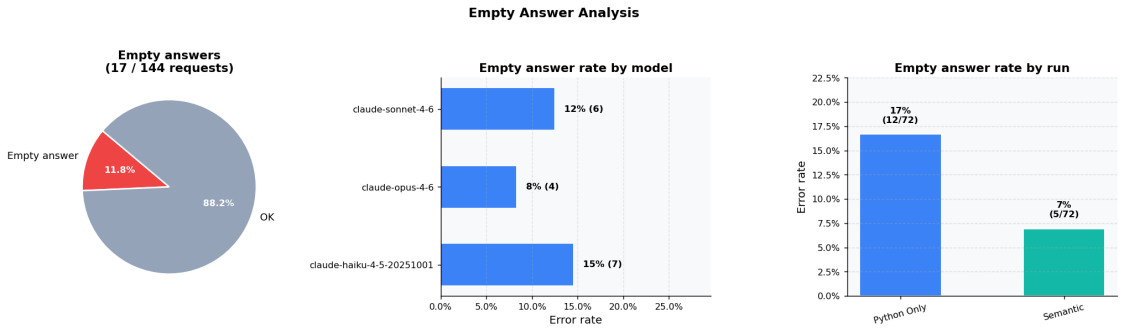
1.6 6 — Accuracy by Injector Category



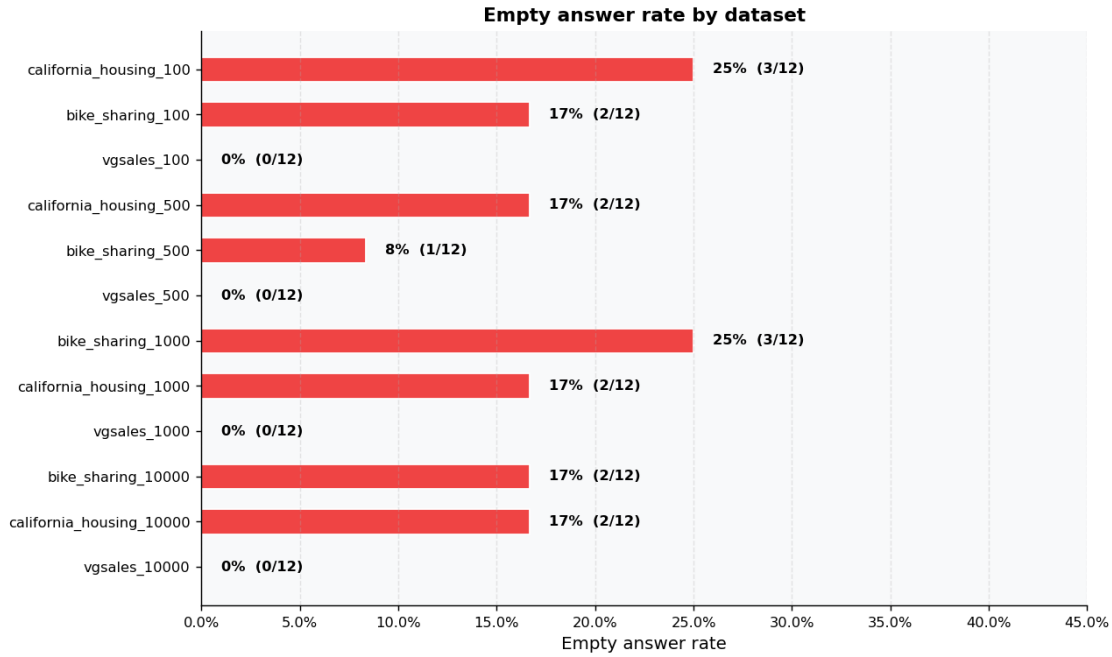
1.7 7 — Empty Answer Analysis

```
Empty answers: 17 / 144 (11.8%)
Affected datasets: ['bike_sharing_100', 'bike_sharing_1000',
```

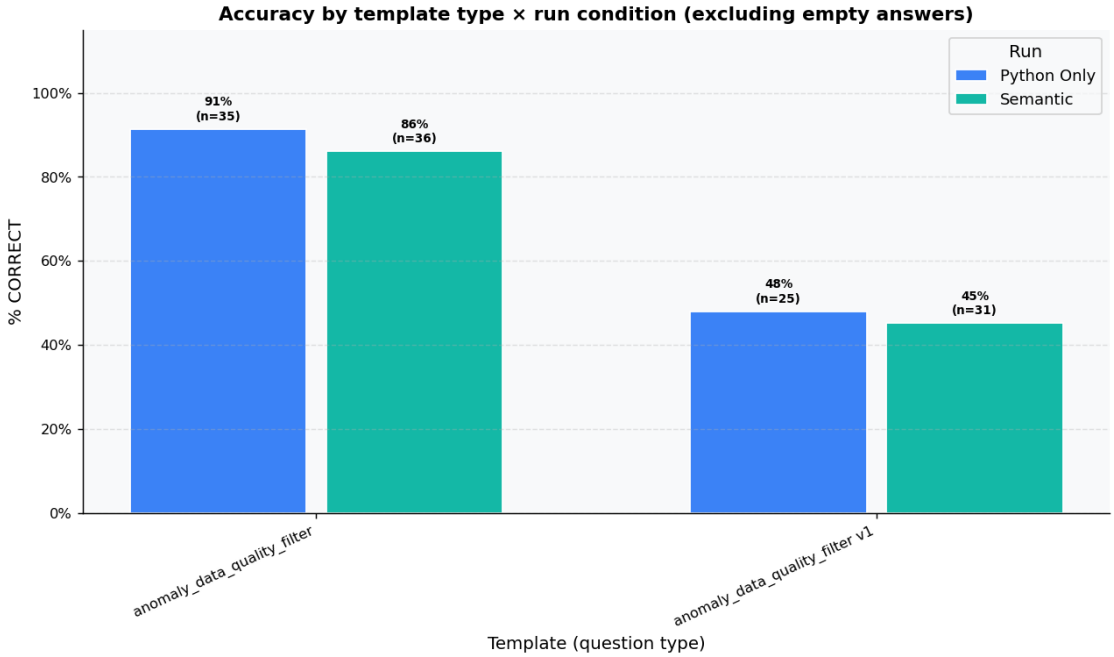
```
'bike_sharing_10000', 'bike_sharing_500', 'california_housing_100',
'california_housing_1000', 'california_housing_10000', 'california_housing_500']
```




```
Empty answers by model × run:
                                        errors  total rate
model                      run
claude-haiku-4-5-20251001 python_only       4     24  17%
                           semantic         3     24  12%
claude-opus-4-6            python_only      3     24  12%
                           semantic         1     24   4%
claude-sonnet-4-6          python_only      5     24  21%
                           semantic         1     24   4%

Empty answers by dataset:

                          errors  total  rate rate_fmt
dataset
california_housing_100         3     12  0.25      25%
bike_sharing_100               2     12  0.17      17%
vgsales_100                    0     12  0.00       0%
california_housing_500         2     12  0.17      17%
bike_sharing_500               1     12  0.08       8%
vgsales_500                    0     12  0.00       0%
bike_sharing_1000              3     12  0.25      25%
california_housing_1000        2     12  0.17      17%
vgsales_1000                   0     12  0.00       0%
bike_sharing_10000             2     12  0.17      17%
california_housing_10000       2     12  0.17      17%
vgsales_10000                  0     12  0.00       0%

Datasets with errors: 8 / 12
```

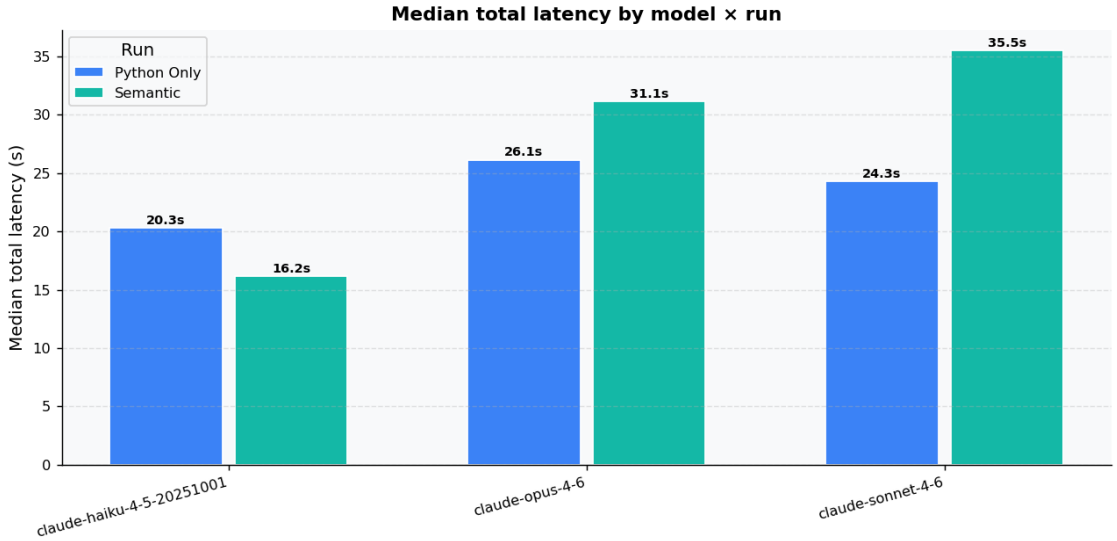


1.8 8 — Template × Run Accuracy

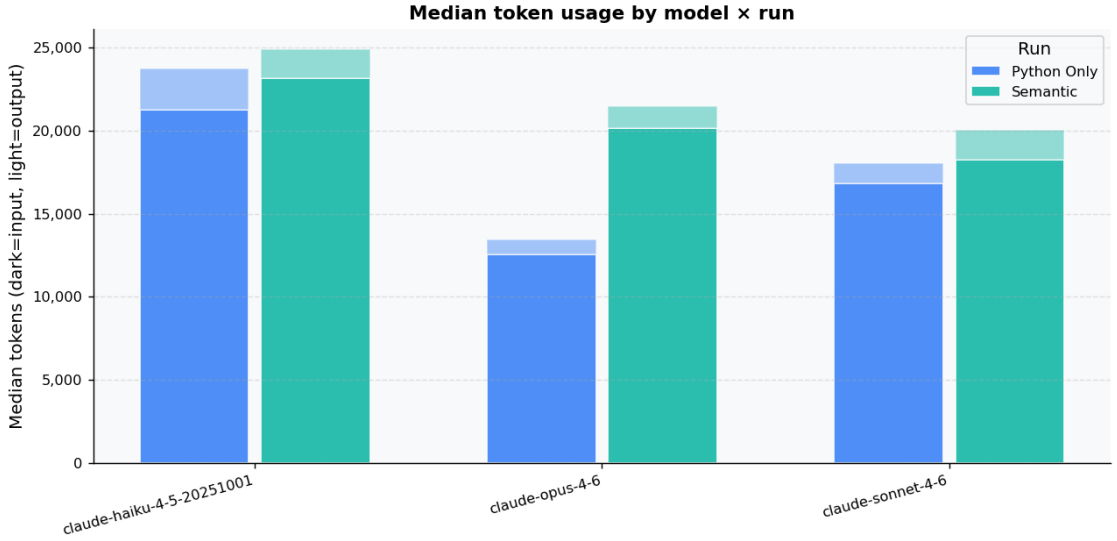


1.9 9 — Latency & Token Metrics

All charts use **median** values (robust to outliers). Error entries are excluded. Metrics: `total_latency_s` (wall-clock), `ttft_s` (time-to-first-token), `input_tokens`, `output_tokens`.



No TTFT data available.



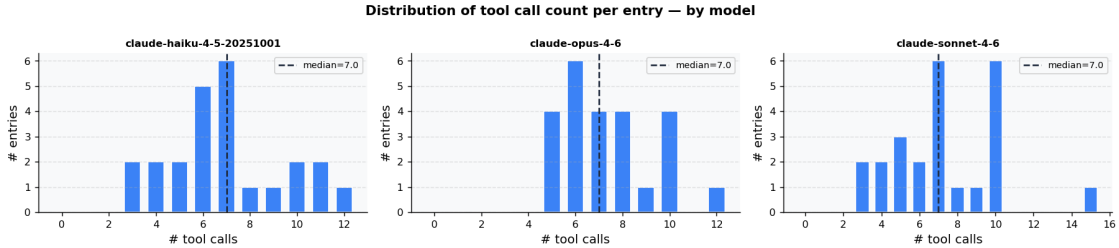
Median metrics per run × model (errors excluded):

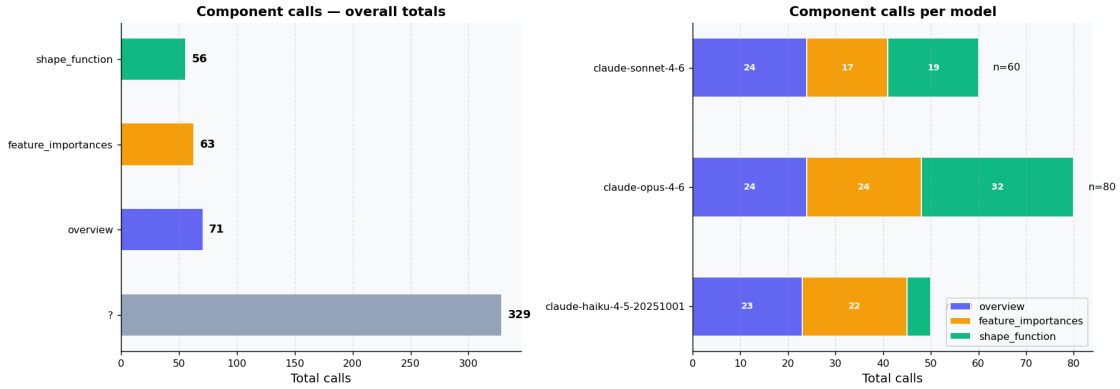
	n	total_latency_s	ttft_s	input_tokens

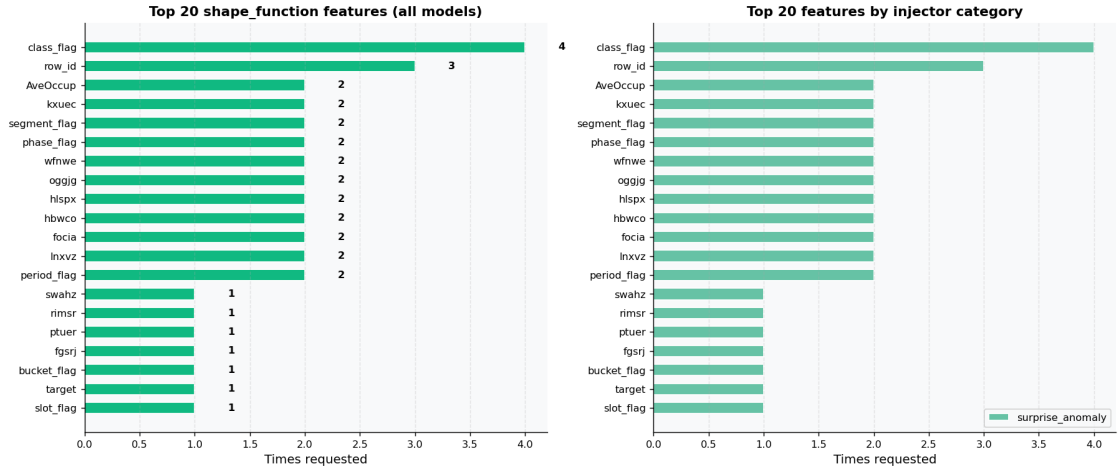
```
output_tokens  total_tokens
run         model
Python Only claude-haiku-4-5-20251001  20            20.29   None      21243.50
2488.00      23664.00
            claude-opus-4-6            21            26.13   None      12543.00
920.00      13463.00
            claude-sonnet-4-6          19            24.26   None      16805.00
1229.00      18214.00
Semantic    claude-haiku-4-5-20251001  21            16.17   None      23147.00
1741.00      24757.00
            claude-opus-4-6            23            31.12   None      20151.00
1308.00      21401.00
            claude-sonnet-4-6          23            35.50   None      18274.00
1762.00      20036.00
```

1.10 10 — Semantic Tool Usage Deep Dive

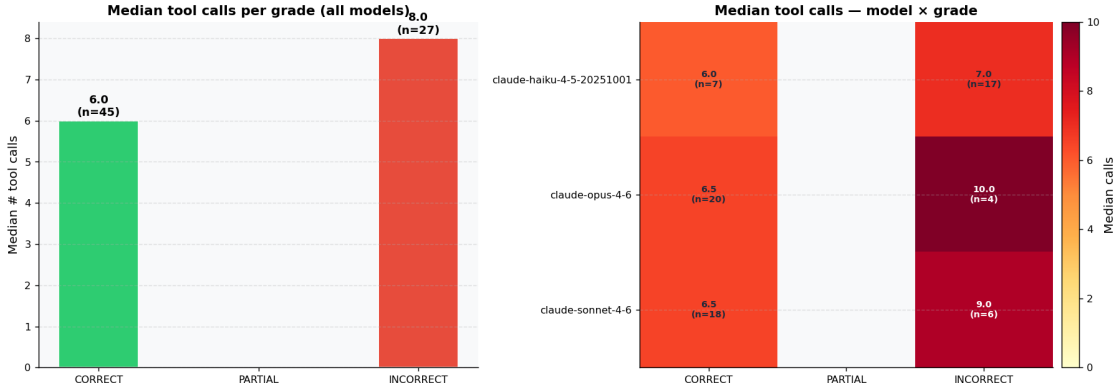
```
Entries in semantic run       : 72
Entries with 0 tool calls     : 0  (0.0%)
Total tool calls made         : 519
Median calls per entry        : 7.0
Max calls per entry           : 15

Calls per entry by model:
                          median  mean  max  zero_calls
model
claude-haiku-4-5-20251001    7.00  6.96   12           0
claude-opus-4-6              7.00  7.38   12           0
claude-sonnet-4-6            7.00  7.29   15           0

Component breakdown (all entries):
component
?                      329
overview                71
feature_importances     63
shape_function          56
```





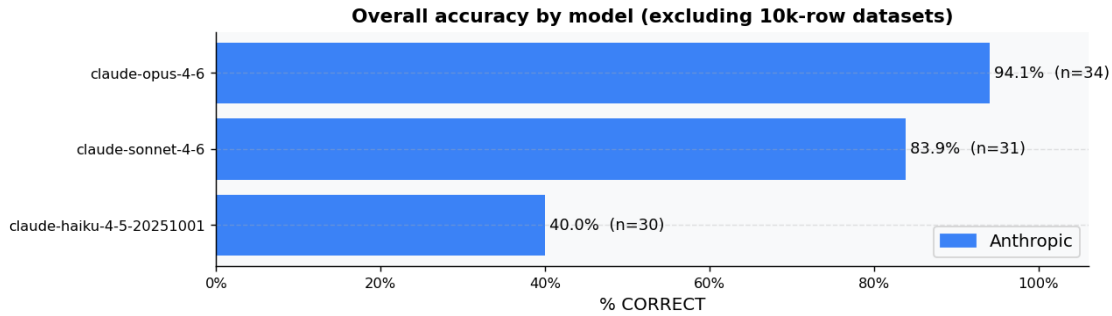
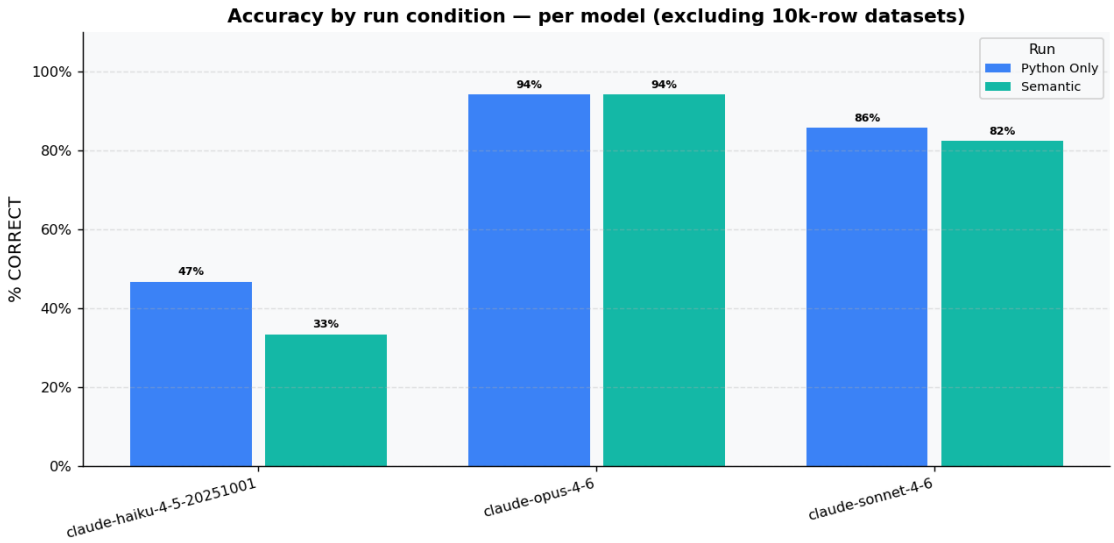


posx and posy should be finite values
posx and posy should be finite values



```
Entries where no tools were called: 0

No zero-call entries found.
```

1.11 11 — Performance Excluding 10k-Row Datasets

Since 10k-row datasets cause empty answers for csv_context (29% error rate), this section filters them out entirely so we can compare modes on an equal footing.

```
Records after excluding 10k-row datasets: 95 / 127
Datasets kept: ['bike_sharing_100', 'bike_sharing_1000', 'bike_sharing_500',
'california_housing_100', 'california_housing_1000', 'california_housing_500',
'vgsales_100', 'vgsales_1000', 'vgsales_500']
```